RA8876_DX_FT8 June 1, 2025

Background
Several years ago (June 2023) I put together a version of our FT 8 code based on the RA8876 Display. The RA887 Display Code and Interface Board were plagiarized from this site:
https://oshwlab.com/k7mdl/v2-1_7-inch_teensy_pcb

Also, the base FT8 Encoding / Decoding Algorithms are based on work by Karlis Goga:
https://github.com/kgoba/ft8_lib

As you will see this version made use of the 1024 X 600 Display to show locations of worked FT8 Stations on a World Map. https://www.buydisplay.com/spi-7-inch-tft-lcd-dislay-module-1024x600-ra8876-optl-touch-screen-panel

For whatever reason I can not remember why I put this work aside.

Now with the success with the DX_FT8 project I have revived my interest in this project.

So I designed and had fabricated a board which works with either a DX-FT8 Five Band or Seven Band Board with the only modification of replacing a user fabricated audio interface cable with a simple three pin header pin for a direct connection.

I was pleased that the five boards that were fabricated by JCLPCB required no modification to work with my DX_FT8 boards.

Here is a view of the current project software taken while I was operating portable in Cornwall, UK

Software Design General
As to Software, I plan for the software to be a close **BUT Not Exact** port of the DX_FT8 project software, V1.9.4. There are several reasons why the software will be different:

1) Using a different processor and display.
2) Using a different IDE (Teensy Arduino IDE has far more usable device drivers than STM IDE);
3) Faster Processor (About 450 MHz).

Software Design Scope
The obvious software addition is the extra free space on the display. As a first approach I have filled it with world maps which shows completed FT8 QSO's. Users may want to use the display area for something else.

A design note will be provided on how a usermay create and maintain their own display map.

I am using the Teensy RTC instead of the DX FT8 Board RTC (DS3231Chip).

Also, I have reverted back to using a GPS receiver to aid in time synchronization. I have found that when you power the DX FT8 board while downloading firmware, the Teensy RTC is automatically set, both date and time. I do not know how or why this happens, but it does. Accordingly, I have not ported over the eight buttons on the Tune screen for date time setting. The GPS receiver will also update the RTC when a GPS Fix is made.

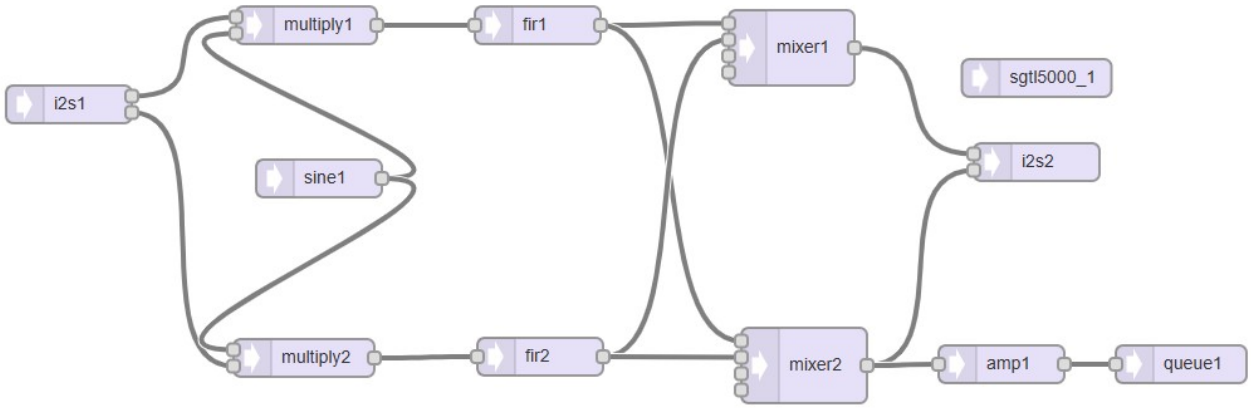
Also, I have added functionality that uses the GPS location. Specifically, as a user moves to a different location the Maidenhead Locator Sent is updated to the new location rather than the user's home location. And, the user's location on the map display will be updated.

Building The Firmware
The firmware is built using the Arduino IDE which has been modified with the Teensy Application called Teensyduino.

The software was developed using the Teensy Audio System Design Tool which may be found here:
https://www.pjrc.com/teensy/gui/?info=AudioRecordQueue

Teensy Audio Design Sketch

Here is the Teensy Audio Design Sketch used for the project firmware.



The Teensy Audio Library is designed to run at an audio sample rate of 44.1 kbps.

However, in order to efficiently do FT8 Demodulation the DX FT8 software was designed to run at an audio sample rate of 32 kbps. So, two of the Teensy Library Files were modified to allow the firmware to run at 32 kbps.

Here is how I installed the Arduino IDE with Teensyduino and then modified two of the Audio Library Files.

Installing and Modifying IDE

First, I downloaded Arduino 1.8.5 which is available here:
https://www.arduino.cc/en/software/OldSoftwareReleases/

After extracting the Arduino 1.8.5 version of the IDE I then downloaded the Teensyduino Application which may be found here: https://www.pjrc.com/teensy/td_download.html

On this site, I paged down to the section shown below and choose the version for my machine.

Arduino 1.8.x Software Development (Download and Run Installer)

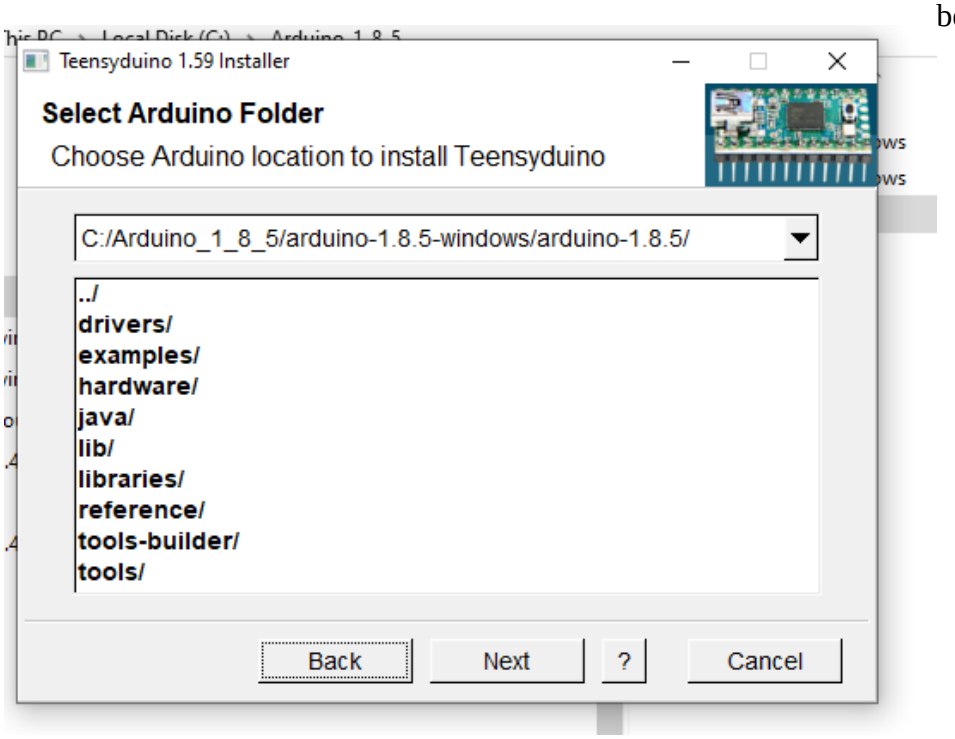
Teensyduino is a software add-on for the Arduino software.

- Macintosh Complete Software
  Supports: Catalina, Big Sur, Monterey, Ventura, Sonoma
- Macintosh Installer for Arduino on Older Macs
  Supports: Mojave
- Linux Installer (X86 32 bit)
- Linux Installer (X86 64 bit)
- Linux Installer (ARM 32 bit / Raspberry Pi)
- Linux Installer (ARM 64 bit / AARCH64 / Jetson TX2)
- Windows XP / 7 / 8 / 10 / 11 Installer

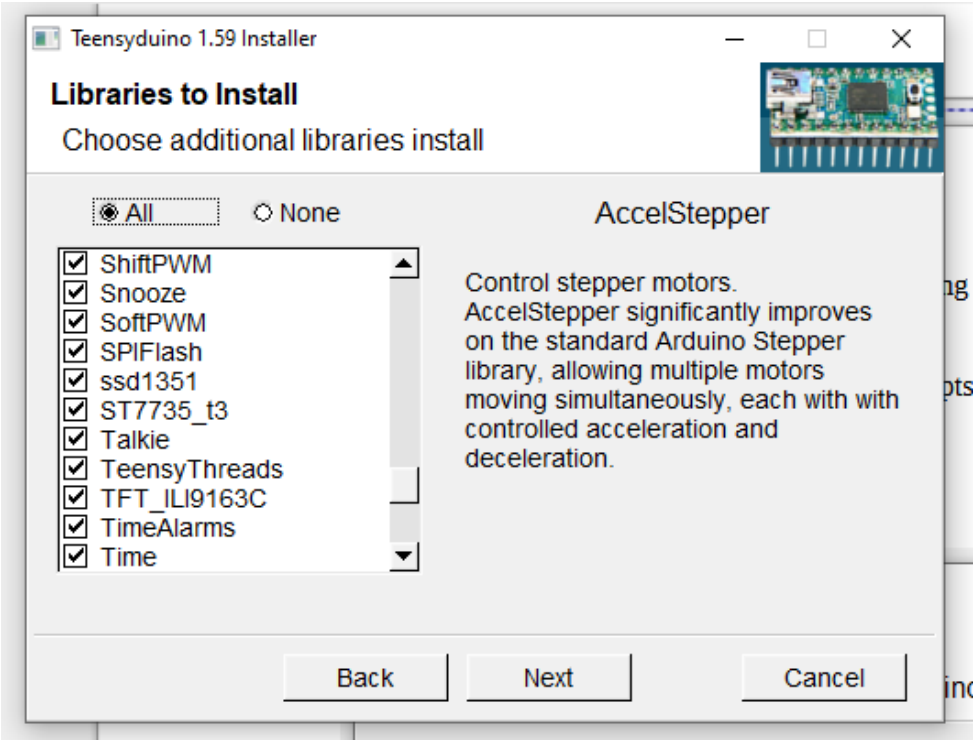
Teensyduino 1.59 supports Arduino versions 1.8.5 and 1.8.9 and 1.8.13 and 1.8.15 and 1.8.16 and 1.8.19.

As stated above I installed Arduino 1.8.5 by simply extracting the IDE from the Archive File to a folder on my hard drive.

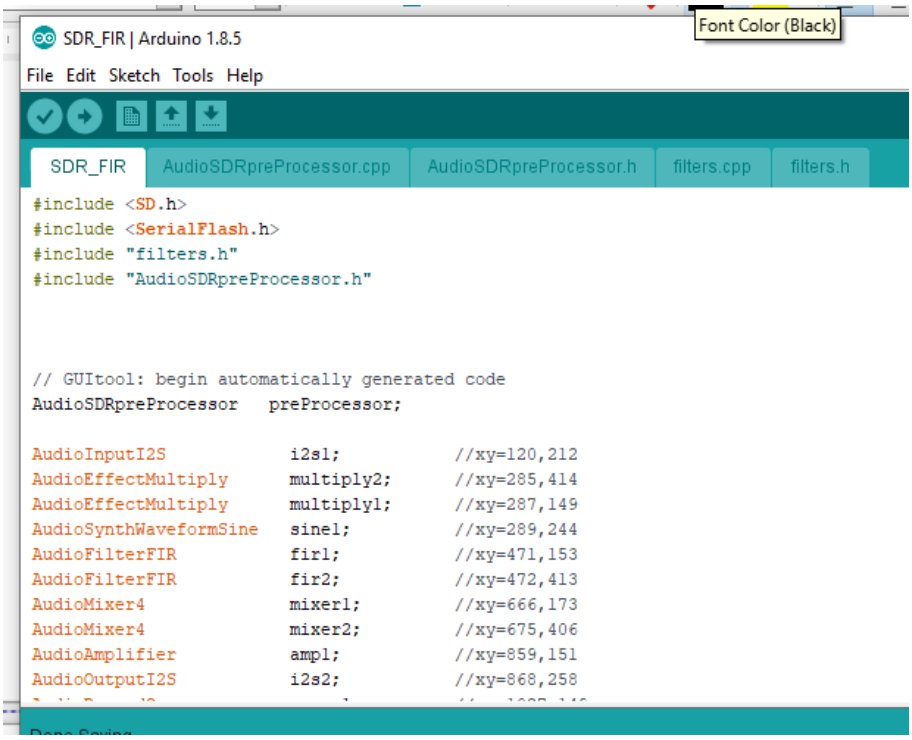
Then I ran the Teensyduino Installer and followed the prompts to arrive at the folder location shown below.



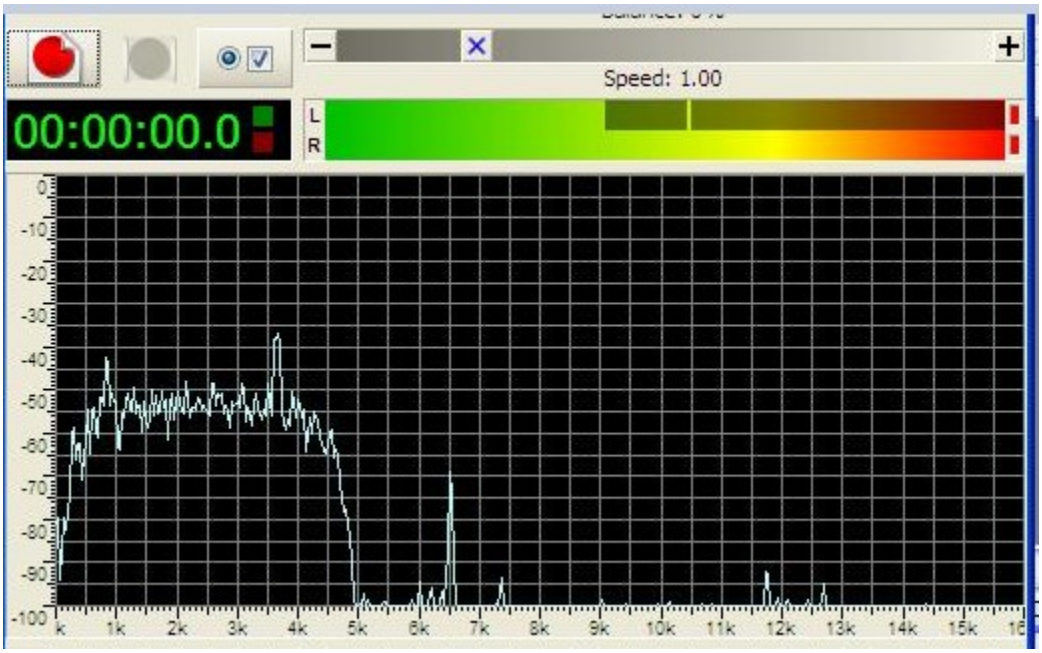
I then accepted installation of all available libraries as shown below:


Teensyduino 1.59 Installer
Libraries to Install
Choose additional libraries install
All
None
AccelStepper
ShiftPWM
Snooze
SoftPWM
SPIFlash
ssd1351
ST7735_t3
Talkie
TeensyThreads
TFT_ILI9163C
TimeAlarms
Time
Control stepper motors.
AccelStepper significantly improves on the standard Arduino Stepper library, allowing multiple motors moving simultaneously, each with with controlled acceleration and deceleration.
Back
Next
Cancel

I then opened the Arduino IDE 1.8.5 and loaded the test file SDR_FIR which implements the Teensy Audio Design Sketch shown above.

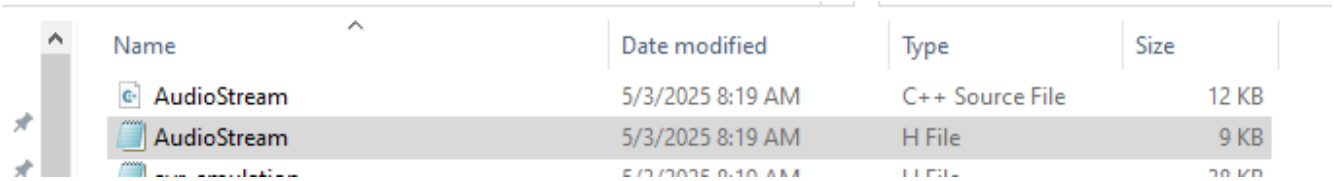


This results in a simple SDR receiver with a received bandwidth of almost 5 kHz.



I then drilled down into the Arduino IDE to locate the AudioStream.h file found in the folder shown below:

C:\Arduino_1_8_5\arduino-1.8.5-windows\arduino-1.8.5\hardware\teensy\avr\cores\teensy4



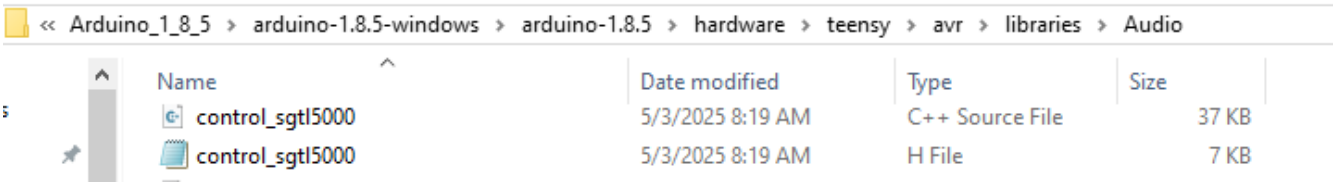
Name	Date modified	Type	Size
AudioStream	5/3/2025 8:19 AM	C++ Source File	12 KB
AudioStream	5/3/2025 8:19 AM	H File	9 KB

I then opened the file and made the modification shown below:

```
#ifndef AUDIO_SAMPLE_RATE_EXACT
#//define AUDIO_SAMPLE_RATE_EXACT 44100.0f
define AUDIO_SAMPLE_RATE_EXACT 32000.0f
```

I then drilled down into the Arduino IDE to locate the control_sgtl5000.cpp file found in the folder below:

C:\Arduino_1_8_5\arduino-1.8.5-windows\arduino-1.8.5\hardware\teensy\avr\libraries\Audio



« Arduino_1_8_5 › arduino-1.8.5-windows › arduino-1.8.5 › hardware › teensy › avr › libraries › Audio

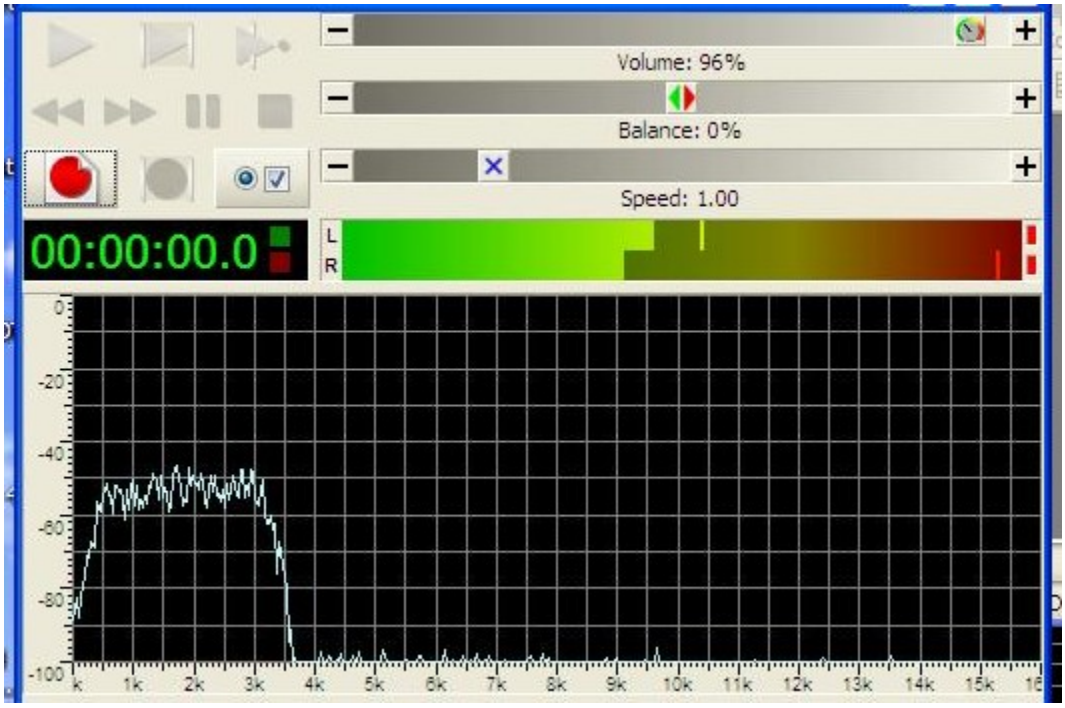
Name	Date modified	Type	Size
control_sgtl5000	5/3/2025 8:19 AM	C++ Source File	37 KB
control_sgtl5000	5/3/2025 8:19 AM	H File	7 KB

I then opened the file and made the modification shown below:

```

//#define CHIP_CLK_CTRL          0x0004
#define CHIP_CLK_CTRL           0x0000
```

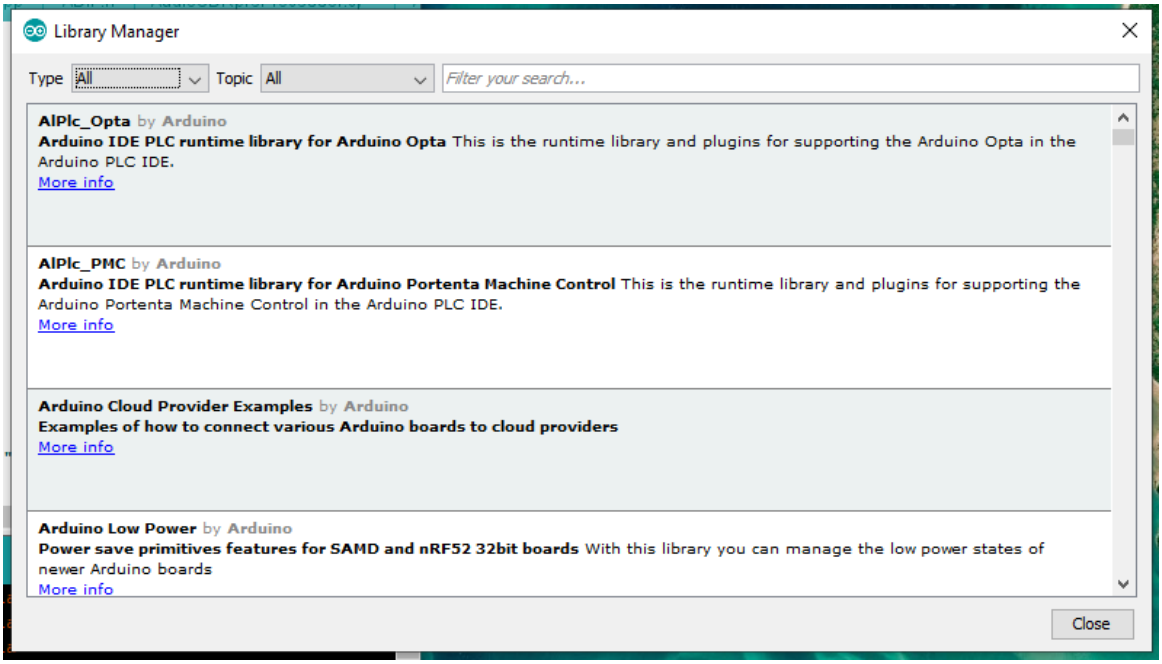
I then reran the SDR_FIR project and now found an audio output restricted to less than 3.5 kHz bandwidth which is required by DX_DT8 Project firmware.

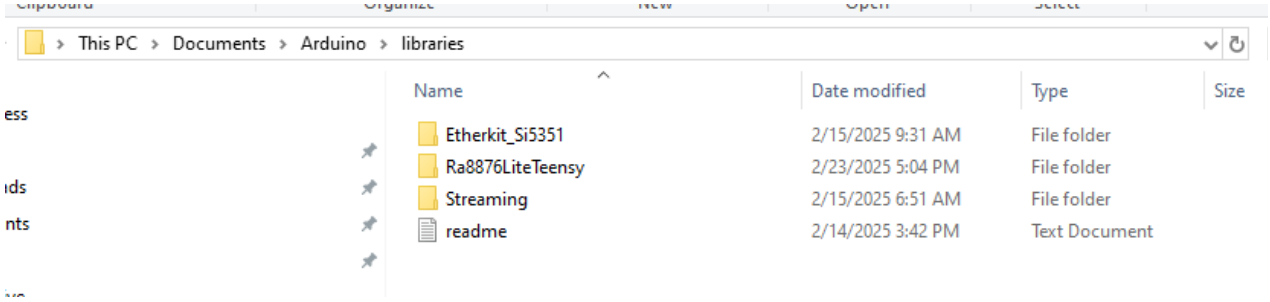


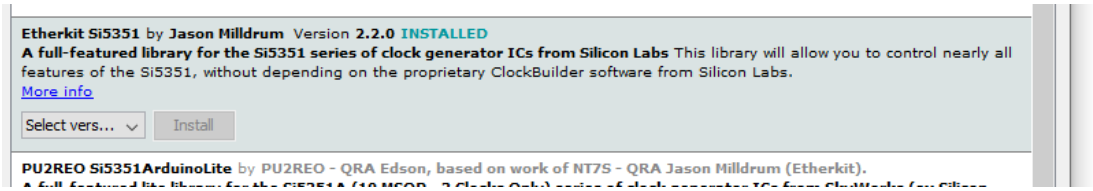
This confirms that the audio sampling is done at 32000 bps instead of 41000 bps.

Library Installation Details

I then installed the three Arduino Libraries as shown below by using the IDE Library Manager







After installing the three libraries listed above I was then able to build the current version of the Firmware as shown below:

All of the other files required for the build are included in the project folder.

Yes, I know that Version 1.8.5 of the Aruduino IDE is quite dated. I installed Version 2.3.4 of the Arduino IDE however even after several days battling with the installation I was not able to get the RA8876 DX FT8 project to run at 32kbps. So I reverted back to V1.8.5 which is still available and works well.

Map Display Notes

At the present the application will display maps at three scales. The image files are embedded in the firmware. Maps are drawn in a polar convention (distance and bearing). So far maps have been made for North America and Northern Europe. The maps are manipulated by Maidenhead Locator. Each Map is centered on a Maidenhead Locator and all stations are plotted by Maidenhead Locator. The North America Map is centered on EM29 while the Northern Europe Map is centered on JM29.

Right now there are maps for EM29 and JM29. The user may select the Region and Map Scale from the Tune Screen.

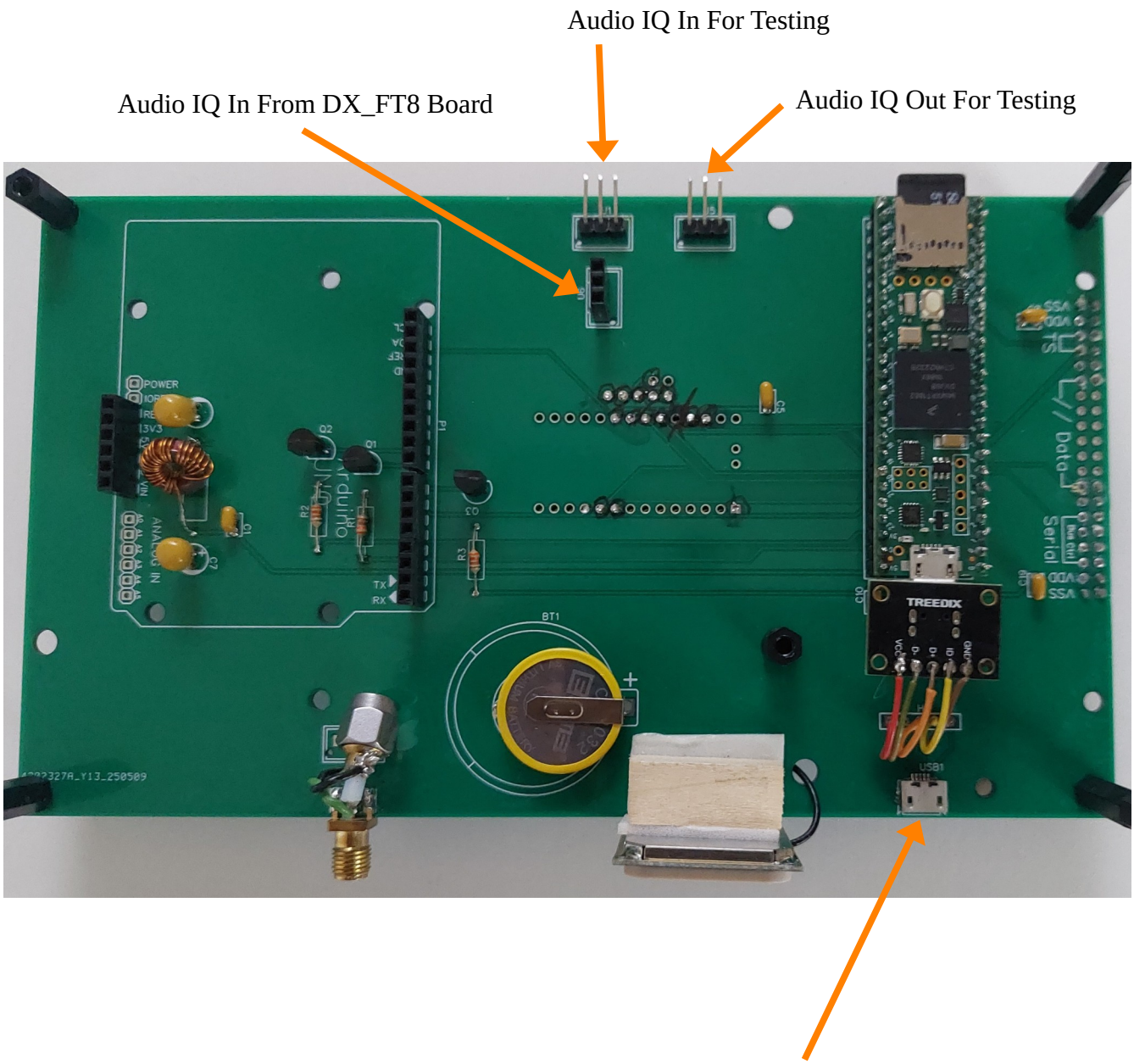
The user station location is shown as a green dot while stations worked are shown as red dots.

Hardware Notes

The project board is designed so that the LCD Display and DX FT8 Board may be directly plugged in without any cables being required.

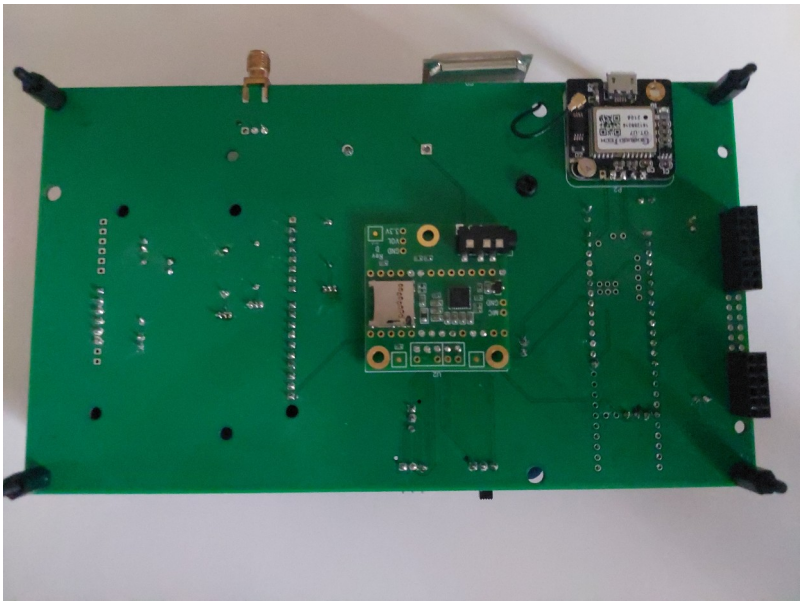
Photos are shown below.

Completed Board Top Side

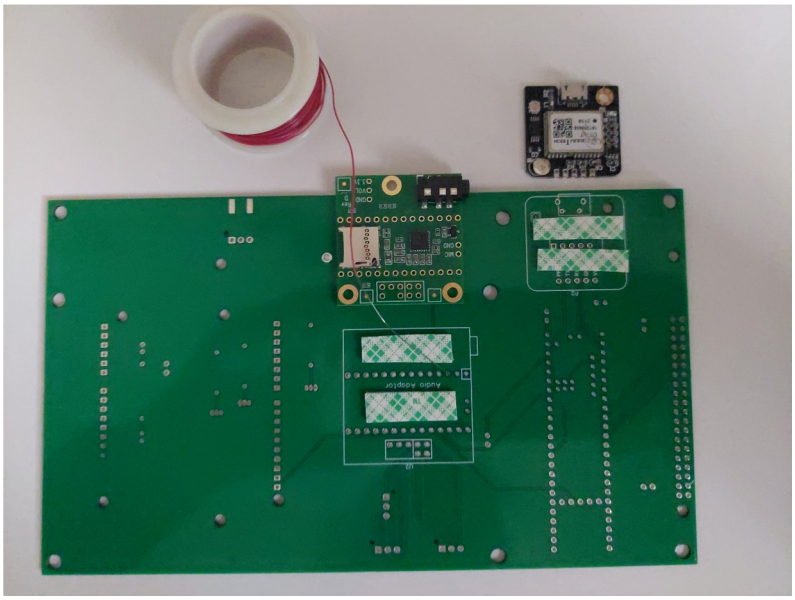


All parts to be installed are through hole parts except for the USB Micro Female Socket shown here.

Completed Board Bottom Side

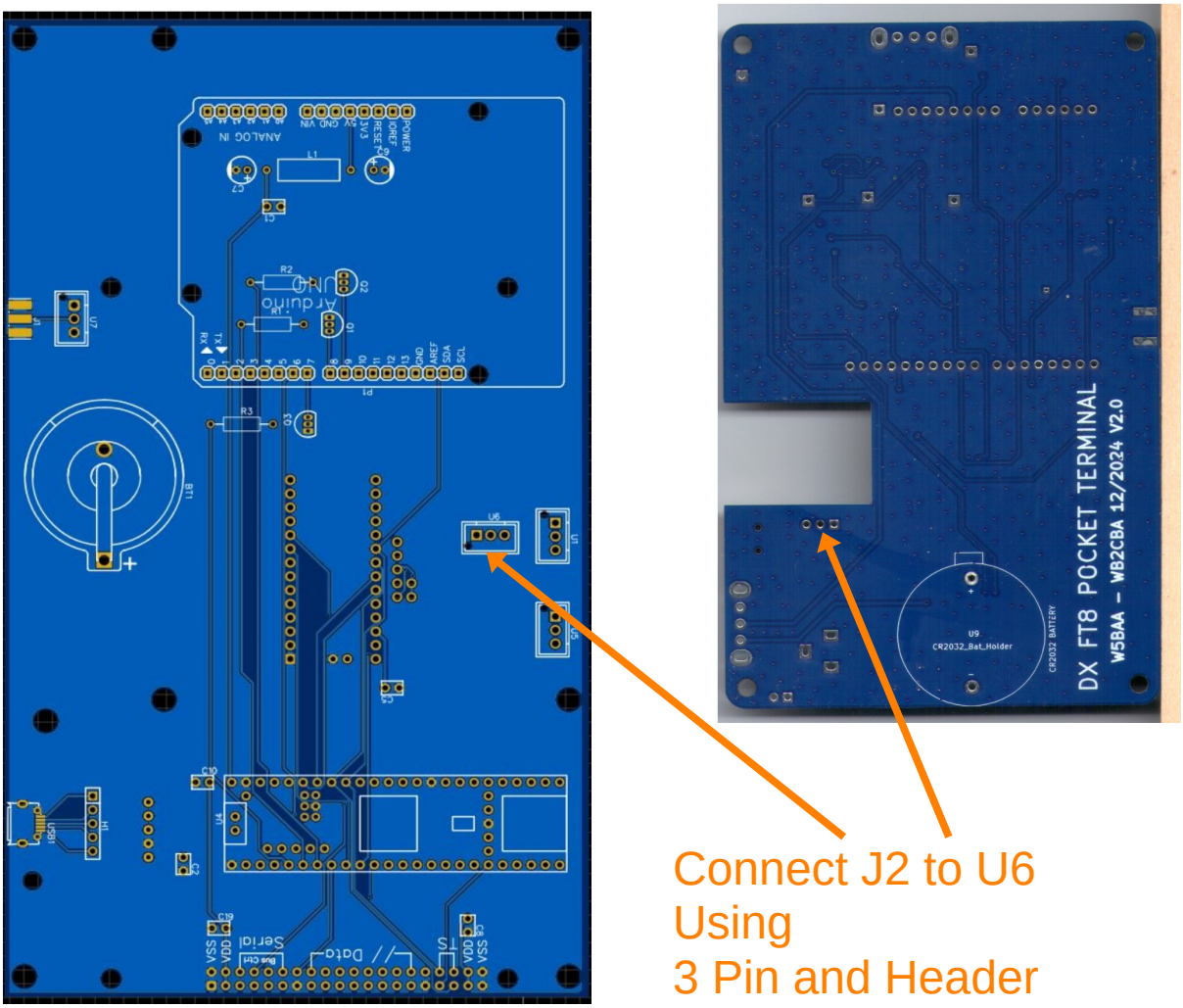


As can be seen the Teensy Audio Shield and GPS Receiver are installed on the Bottom Side of the Board.



In order to minimize the height of the installed boards they boards were directly soldered in with out pins or headers using double sided tape along with thin solid wire wrap wire. This makes for a compact installation. And, if you wish to remove the board it is quite easy to do so with small nail scissors and and Exacto Kinfe.

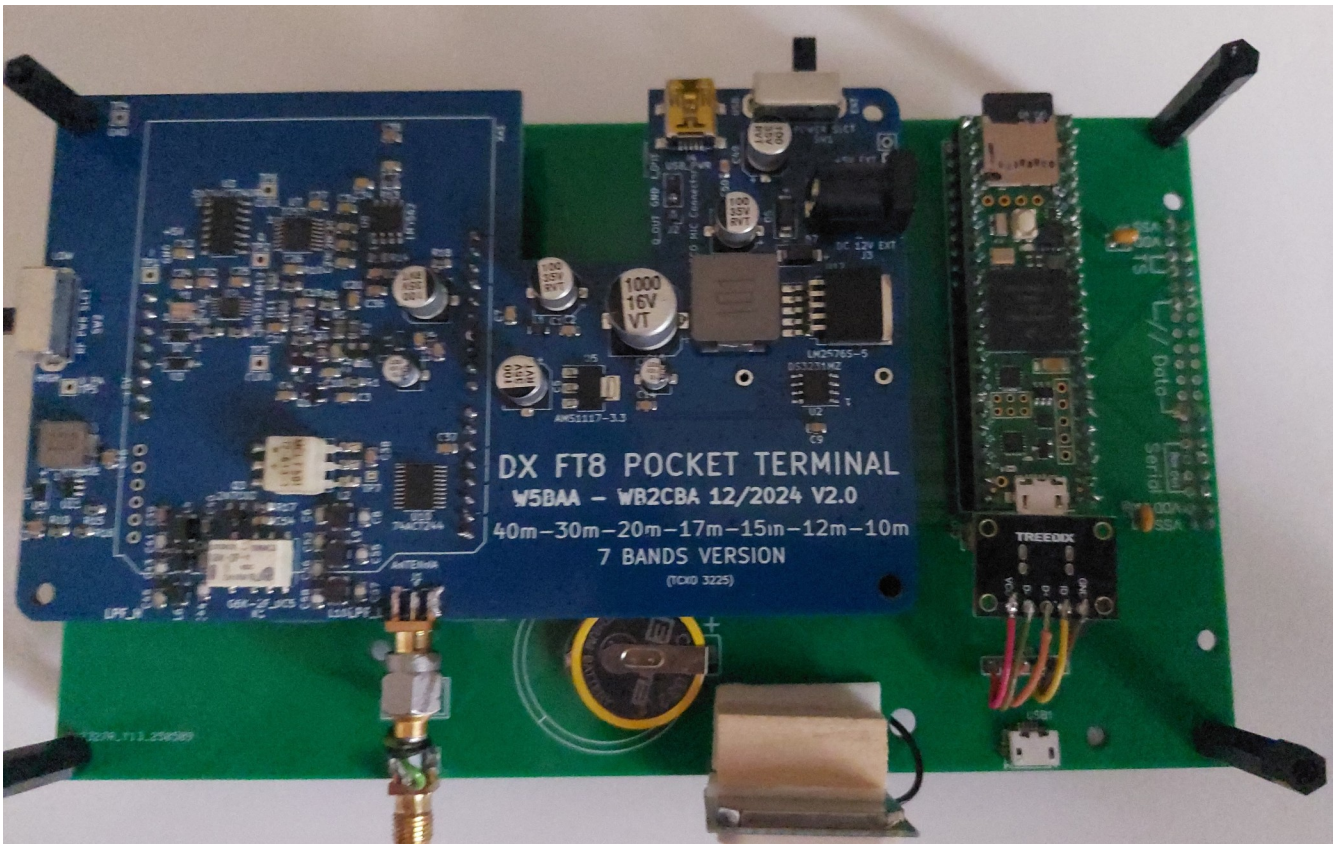
Here is a drawing which details the Audio IQ connection between the RA8876 Board and DX FT8 Board.



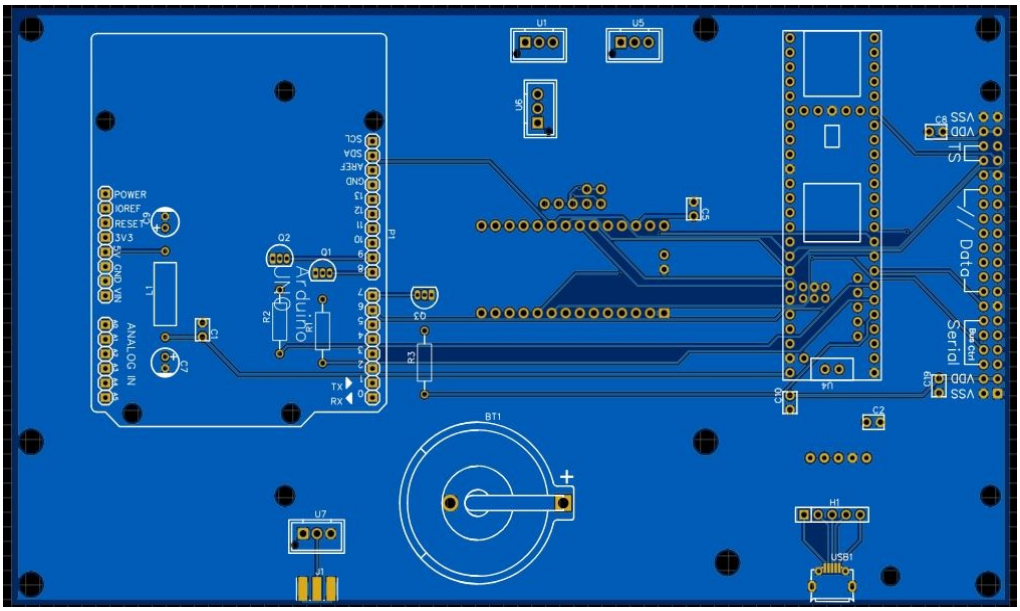
Power for the entire package is provided thru the DX FT8 Board. Power may be supplied using either the 12 VDC Port or the USB Mini connector on the DX FT8 Board.

When laying out the RA8876 DX FT8 Board I had intended to do stitching between the top and bottom side but I failed to do so before placing the Fabrication Order. In usage, I have found the Audio IQ Spectrum to be quite clean for when displayed as the Waterfall.

Completed Board With DX_FT8 Plugged In

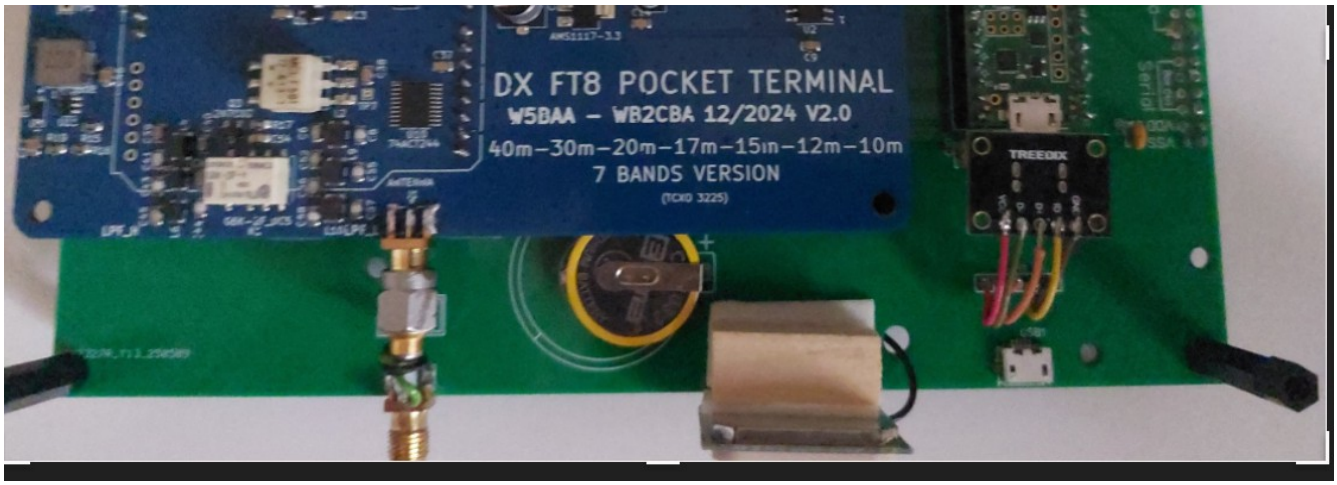


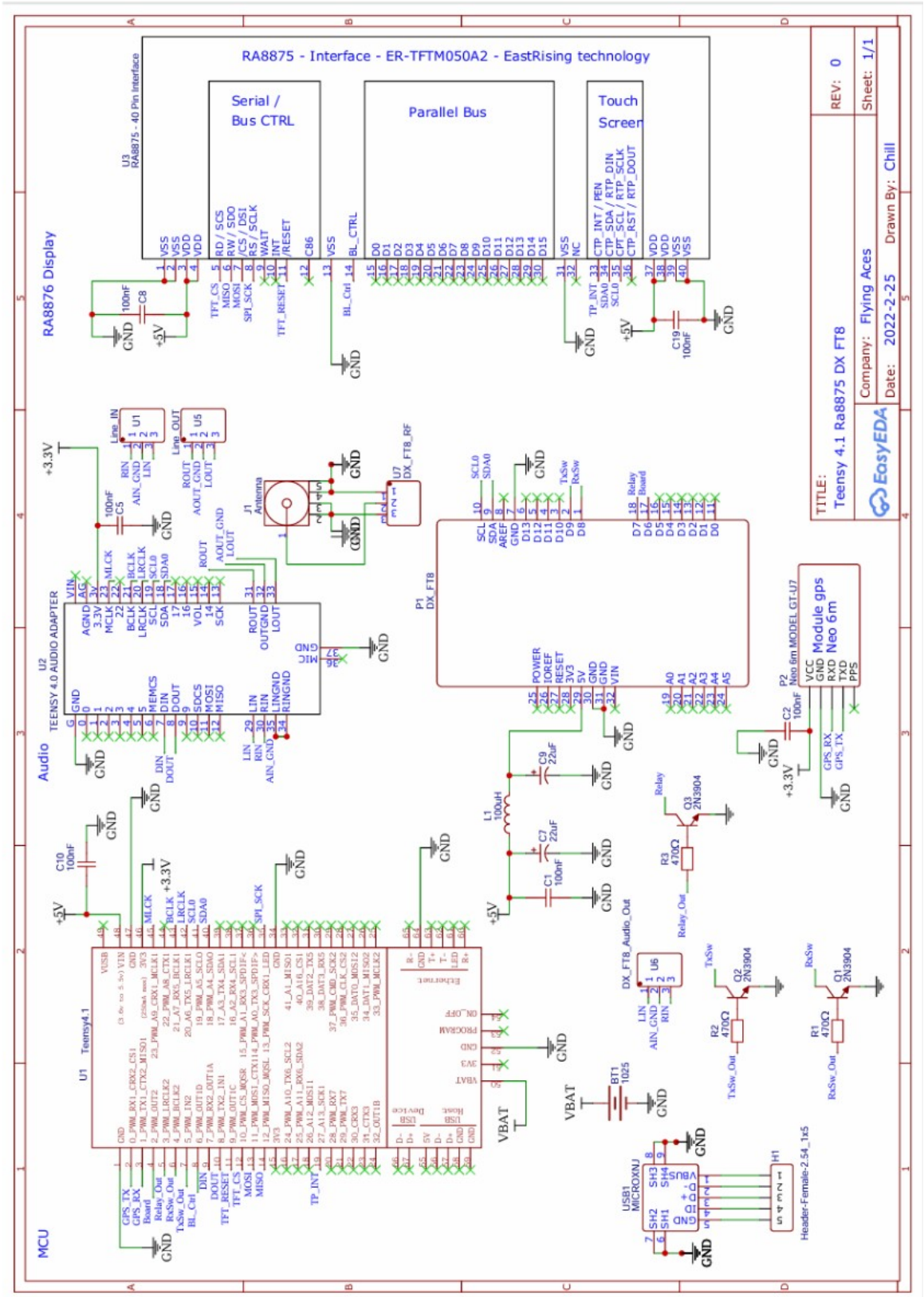
Edge Connectors



In order to make for easier access to the RF Antenna SMA Connector and the Teensy 4.1 USB Port edge connectors have been provided at the top edge of the board.

The photo below illustrates the connections made between the DXFT8 Board and the Teensy 4.1 and the project board. The USB edge connector has SMD connections which are fairly easy to make. If you do not want to do an SMD connection you can use a USB Micro Breakout board installed with double sided tap. A USB Micro Male Connector Breakout board is plugged into the Teensy 4.1 USB Port.



Use of these edge connectors is not required for full and proper use of the RA8876 DX FT Board.

Board Schematic